

Praktikum Hochfrequenz-Schaltungstechnik WS25/26

Bericht

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5.5.1

Laut dem Datenblatt zum CMOS-Sechsfach-Inverter 74AC14 [1] hat der Pulsgenerator eine Anstiegszeit von $t_{\text{rise}} = 3 \text{ ns}$ and a fall time of $t_{\text{fall}} = 3 \text{ ns}$. According to the approximation formula of the rise time of DSOs

$$t_o = \frac{0.35}{BW}$$

the rise time for a 60 MHz (120 MHz) DSO are $t_o = 5.8 \text{ ns}$ ($t_o = 2.9 \text{ ns}$) [1]. Thus the DSO cannot resolve rise times lower than those calculated values. Since the rise and fall times are significantly faster the given DSOs are not able to measure them correctly. This can be confirmed by a measurement with a 60 MHz where the rise time was noted at 5.5 ns and the fall time was 4.5 ns being much higher than the theoretical value.

5.5.2

The formula for reflection coefficients is stated as

$$\Gamma = \frac{Z_t - Z_0}{Z_t + Z_0}$$

where Z_t is the termination impedance and $Z_0 = 50 \Omega$ is the impedance of the wire. For different termination impedances the theoretical reflection coefficients $\Gamma_{\text{theo.}}$ are calculated in table ??.

Table 1: theoretical and measured reflection coefficients at a wire impedance of $Z_0 = 50 \Omega$

Z_t	50Ω	75Ω	93Ω
$\Gamma_{\text{theo.}}$	0	0.20	0.30
Γ_{measured}			

Differences can occur due to losses in the wire since the isolation between inner wire and cylindrical conductor deafens the signal due to its magnetic properties.

5.5.3

The cable lengths were determined from the measured time difference between the incident and the reflected pulse. Since the signal propagates to the cable end and back, the cable length is given by

$$l = \frac{c \cdot k \cdot \Delta t}{2}$$

where c is the speed of light and k is the velocity factor of the cable. The time difference Δt was obtained from the oscilloscope display using the number of divisions and the selected time base.

- Measure the time difference Δt between the incident and reflected pulse on the oscilloscope.
- Convert the oscilloscope reading from divisions to time:

$$\Delta t = N_{\text{Div}} \cdot t_{\text{Div}}$$

- Take measurements for both open and short-circuited cable ends.
- Calculate the mean value to reduce measurement uncertainty:

$$\Delta t_{\text{mean}} = \frac{\Delta t_{\text{open}} + \Delta t_{\text{short}}}{2}$$

- Determine the cable length using the cable's velocity factor k :

Cable type	Δt [ns]	k	calc. Length l [m]	Measured Length l [m]
RG58	38.5	0.66	3.8	3.59
RG188	49.0	0.70	5.1	4.52
SAT (white)	30.5	0.82	3.75	

Table 2: Cable lengths calculated from propagation delay measurements

The results obtained for open and short-circuited cable terminations agree within the measurement uncertainty. Remaining deviations are mainly caused by reading errors, additional cable lengths due to T-connectors.

References

- [1] Axel Bangert. *Praktikum Hochfrequenz-Schaltungstechnik*. Kassel.